

AD 502 – Machine Learning

Pre-Reading Material – Lecture 7

Data Visualization

Estimated Reading Time: 25–30 minutes

Why This Topic Matters

Machine learning involves working with large amounts of data, making it difficult to identify patterns by looking at raw values alone. **Data visualization** converts data into graphical forms that help us understand distributions, relationships, trends, and unusual observations before building a model.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain the purpose of **data visualization** in machine learning.
- Identify appropriate plots for different types of data.
- Understand the use of **Matplotlib** and **Seaborn** for visualization.
- Interpret common plots such as histograms, scatter plots, box plots, and bar charts.
- Identify trends, relationships, distributions, and outliers using visualizations.

Activate Your Prior Knowledge

1. Numbers vs Graphs

Suppose a dataset contains thousands of values representing students' marks.

Think: Would a table of numbers or a graph make it easier to identify the overall performance pattern?

2. Relationship Between Variables

Suppose you have students' study hours and examination marks.

Think: How could you determine visually whether students who study more generally obtain higher marks?

3. Unusual Observations

Imagine that most students score between 50 and 80 marks, but one student scores 5.

Think: Which type of visualization could help you quickly identify this unusual observation?

Prerequisite Concepts

You should be familiar with:

- Python basics
- NumPy arrays
- Pandas DataFrames
- Basic descriptive statistics
- Variables and data types

Core Concepts

1. Data Visualization

Data visualization is the graphical representation of data using charts and plots. It helps identify patterns, trends, relationships, distributions, and unusual observations.

2. Why Visualization is Important in ML

Visualization helps understand a dataset before applying machine learning algorithms. It can reveal data quality problems, outliers, correlations, and patterns that may not be obvious from numerical summaries.



3. Matplotlib

Matplotlib is a widely used Python library for creating static visualizations. It provides functions for producing line plots, scatter plots, bar charts, histograms, and many other graphs.

4. Seaborn

Seaborn is a Python visualization library built on Matplotlib. It provides a simpler interface for creating attractive statistical visualizations and works particularly well with Pandas DataFrames.

5. Line Plot

A line plot connects data points using lines and is useful for showing trends over an ordered variable such as time. It is commonly used for time-series data.

6. Bar Chart

A bar chart represents values for different categories using rectangular bars. It is useful for comparing quantities across discrete categories.

7. Histogram

A histogram shows the distribution of numerical data by grouping values into intervals called **bins**. It helps identify whether data is concentrated, spread out, skewed, or approximately symmetric.



8. Scatter Plot

A scatter plot displays individual observations using points on two numerical axes. It is useful for examining the relationship between two variables.

9. Box Plot

A box plot summarizes the distribution of numerical data using the median, quartiles, and potential outliers. It is useful for comparing distributions and detecting unusual observations.

10. Correlation Heatmap

A correlation heatmap represents correlation coefficients between variables using different visual intensities. It helps identify variables that have strong positive or negative relationships.

11. Distribution

A distribution describes how values of a variable are spread across different ranges. Visualization helps identify characteristics such as concentration, spread, skewness, and possible outliers.

12. Outliers

An outlier is an observation that differs substantially from the other observations in a dataset. Visualizations such as box plots and scatter plots can help identify potential outliers.



13. Univariate Visualization

Univariate visualization focuses on a **single variable**. Histograms, box plots, and count plots are commonly used to understand the distribution or frequency of one variable.

14. Bivariate Visualization

Bivariate visualization examines **two variables simultaneously**. Scatter plots and grouped bar charts can help identify relationships or differences between variables.

15. Multivariate Visualization

Multivariate visualization examines **three or more variables** at the same time. Heatmaps, pair plots, and plots using different colors or sizes can help explore complex relationships.

Guided Reading Questions

1. Why is visualization important before building a machine learning model?
2. When would you use a histogram instead of a bar chart?
3. What type of relationship can be studied using a scatter plot?
4. How can a box plot help identify outliers?
5. What information can be obtained from a correlation heatmap?
6. What is the difference between univariate and bivariate visualization?
7. Why are Matplotlib and Seaborn commonly used in machine learning?

Reflection Activity

Write 100–150 words explaining how data visualization can help a machine learning practitioner before model training. Consider how visualizations can reveal **data distributions, relationships, outliers, and data quality issues**.



Self-Assessment

1. Data visualization represents data using _____ forms.
2. Which Python library is widely used for creating plots and charts?
 - A. NumPy
 - B. Matplotlib
 - C. Scikit-learn
 - D. TensorFlow
3. Which plot is most suitable for showing the distribution of a numerical variable?
 - A. Histogram
 - B. Pie chart
 - C. Line plot
 - D. Network graph
4. Which plot is commonly used to study the relationship between two numerical variables?
 - A. Histogram
 - B. Scatter plot
 - C. Bar chart
 - D. Box plot
5. Which visualization is particularly useful for detecting potential outliers?
 - A. Box plot
 - B. Line plot
 - C. Pie chart
 - D. Area chart
6. A correlation heatmap is primarily used to visualize _____ between variables.
7. Visualization involving a single variable is called _____ visualization.
8. A histogram divides numerical values into intervals called _____.
9. Seaborn is built on top of _____.
10. Data visualization should ideally be performed before _____ a machine learning model.

Key Takeaway

Data visualization is an essential step in the machine learning workflow because it allows us to **see the data before modeling it**. Appropriate visualizations help identify distributions, relationships, trends, and outliers, leading to better understanding and more informed machine learning decisions.

